UNDERGRADUATE RESEARCHER'S COMPASS:

“A PRACTICAL GUIDE FOR FINAL YEAR COMPUTER SCIENCE STUDENTS”

Mobile and Web based FYPs

FACULTY OF COMPUTER AND INFORMATION TECHNOLOGY

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* FYP (Final Year Project)

INTRODUCTION

This guideline is designed to help undergraduate Computer Science students at Jamhuriya University, and it helps in the areas of Computer Applications, Computer Multimedia, and Computer Networks & Security plan, carry out, and present their Final Year Projects (FYPs). It explains the expectations for research quality, ethical practice, technical work, and report writing, while also recognizing the specific methods and evaluation needs of each specialization. Students are expected to produce a working project or study, write a thesis, submit a paper and present their work in a defense.

The guideline applies to different types of projects such as Artificial Intelligence, software prototypes, machine learning, Internet of Things, multimedia systems, human computer interaction studies, network simulations, and security experiments. It provides directions on how to choose a topic, follow the right methodology, implement and test the project, analyze and discuss results, and finally prepare conclusions, formatting, and defense. The key principles are originality, honesty, reproducibility, practical awareness, and clear communication.

Purpose of the guideline:

The purpose of this guideline is to provide clear directions for students as they complete their Final Year Project (FYP). It sets a common standard across all computer science specializations so that every project is assessed fairly and consistently. It explains what students need to deliver, including the proposal, final thesis, project demonstration, presentation defense and paper submission to internationally respected and indexed publishers. The guideline also helps students choose suitable research titles, follow ethical practices, and use proper referencing.

CHOOSING RESEARCH TOPIC

A good research title should be **clear, specific, and focused**. It should tell the reader what the project is about without being too long or too vague. A title should include the main subject, the method or approach, and the target problem or area of application. Titles should not be just one or two words, like "Networks" or "Applications," because these are too broad. Instead, they should describe the actual research being done. For example, instead of writing *"E-Learning System"*, a better title would be *"Design and Implementation of an E-Learning System for Somali Universities in Mogadishu."*

Weak titles are usually too short, too broad, or unclear, while strong titles show **what is built, why it is useful, and where it applies**, Examples are:

Weak Title (too broad): "Computer Security" **Better Title:** "Enhancing Online Banking Security Using Two-Factor Authentication: A Case Study of Mogadishu Banks"	**Weak Title (too general):** "Hospital System" **Better Title:** "Development of a Web-Based Patient Management System for Benadir Hospital in Mogadishu"	**Weak Title (unclear):** "Multimedia in Security" **Better Title:** "Development of the decentralized biometric identity verification system using blockchain technology and computer vision"

You can make broad topic into specific by consulting with your supervisor, and many independent students usually start with a broad idea, like "Mobile Applications" or "Cybersecurity." The key is to narrow it down to something researchable and practical. Start with the general area, then add the purpose, method, and context. This step helps students move from an idea to a researchable title that can be managed within the FYP timeframe, Examples are:

General Topic: Mobile Applications **Specific Title:** "Development of a Mobile Application for Public Transport Scheduling in Mogadishu"	**General Topic:** Cybersecurity **Specific Title:** "Detection of Phishing Websites Using Machine Learning Techniques: A Study in Mogadishu Internet Service Providers"	**General Topic:** Multimedia in agriculture **Specific Title:** "Smart agriculture drone for crop spraying using image-processing and machine learning techniques"

Alignment with Specialization is an important factor so that every student must make sure their topic fits their **specialization**:

- **Computer Applications:** Focus on software development, mobile apps, databases, or business systems. For example: "***Web-Based Inventory Management System for Small Shops in Bakara Market, Mogadishu.***"
- **Computer Multimedia:** Focus on computer vision, graphics, animation, image processing virtual reality, interactive systems, or user experience. For example: "***A Framework for Internet Protocol Based Realtime Bottle Inspection System Using Blobanalysis Technique***."
- **Computer Networks & Security:** Focus on networking, cybersecurity, simulations, or secure communication. For example: "***Simulation and Performance Evaluation of IPv6 Deployment in Mogadishu Internet Networks.***"

SMART AGRICULTURE DRONE FOR CROP SPRAYING USING IMAGE-PROCESSING AND MACHINE LEARNING TECHNIQUES

ABDULLAHI MOHAMED IBRAHIM
MUSTAF HUSSEIN ABDI
MOHAMED MUHIDIN MOHAMUD
ABDIKARIM AHMED OSMAN

SUBMISSION OF GRADUATION PROJECT FOR PARTIAL FULFILLMENT OF THE DEGREE OF BACHELOR OF SCIENCE IN [COMPUTER APPLICATIONS]

JAMHURIYA UNIVERSITY OF SCIENCE AND TECHNOLOGY (JUST)
FACULTY OF COMPUTER & INFORMATION TECHNOLOGY

AUGUST 2026

JAMHRURIYA UNIVERSITY OF SCIENCE AN TECHNOLOGY (JUST)
ORIGINAL LITERARY WORK DECLARATION

Name of Candidate 1: **ID No:**
Name of Candidate 2: **ID No:**
Name of Candidate 3: **ID No:**
Name of Candidate 4: **ID No:**

Name of Degree: Bachelor Degree of Computer Application
Title of Project/Research Report/Thesis:
Field of Study:
We, the undersigned, do solemnly and sincerely declare that:
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Candidate 4's Signature: _________________________ Date: _____ /_____/______
Subscribed and solemnly declared before,
Supervisor's Name: ___________________________ Date: _____/_____/____________
Supervisor's Signature: _________________ Designation: ________________________

ABSTRACT

Definition

An abstract is a summary of a research paper, thesis, or article. It provides readers with a quick overview of the study, including the background, problem, objectives, methods, key results, and conclusions. Its purpose is to help readers understand the essence of the research without reading the full document, allowing them to decide whether the study is relevant to their interests. In academic writing, the abstract is often the first section reviewers and readers look at, making it an essential part of any scholarly work.

Components of an Abstract

1. **Background / Introduction**
   This section provides the context for the study. It briefly introduces the research area, highlights why the topic is important, and establishes the foundation for the research. It often includes a concise statement of the broader issue being studied.
2. **Problem / Gap**
   Here, the abstract identifies the specific problem that motivated the study. It highlights what is missing in existing literature or practice (the gap) and clarifies why research is necessary.
3. **Objective**
   This part states the main aim of the study, usually beginning with phrases like "The objective of this study was…" or "This research aims to…". It is a direct statement of purpose.
4. **Methods**
   This section summarizes how the study was conducted. It includes information about research design (e.g., cross-sectional, experimental), participants or data sources, and key techniques/tools used for analysis.
5. **Results**
   This presents the most important findings of the study in a concise way, usually with statistics or key outcomes. It answers the research question and provides evidence for the conclusions.
6. **Conclusion / Implications**
   The final part interprets the results and explains their significance. It highlights the contribution of the study, its practical or theoretical implications, and sometimes recommendations for future research.
7. ***Keywords***
   A list of important terms (3–6) that represent the main themes of the research, helping with indexing and searchability.

As per the Faculty of Computer and IT, Jamhuriya University of Science and Technology guidelines, the abstract of a **research paper or thesis** must be written in **250-300 words** and presented as a single paragraph. It should be concise, clear, and self-contained, providing readers with a quick overview of the study. The abstract must cover the essential components: **Background/Introduction, Problem/Gap, Objective, Methods, Results, and Conclusion/Implications**. Students are required to use simple and direct language, avoiding unnecessary details, tables, figures, or references. At the end of the abstract, **3–6 keywords** must be included to reflect the main themes of the research and enhance its visibility in academic searches.

***NOTE**: the **abstract** should not be included: references, charts, graphics, figures, maps, etc.*

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Sample

ABSTRACT

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The rapid expansion of e-commerce, coupled with growing urban populations, has intensified the need for efficient, transparent, and accessible delivery services. Despite this demand, conventional delivery systems continue to face challenges such as limited real-time tracking, inefficient routing, delays, and reduced customer satisfaction. Addressing these limitations requires the integration of advanced technologies capable of enhancing both efficiency and user experience. This study presents the development and design of a mobile application for delivery services integrated with Global Positioning Radio System (GPRS) technology. The system was implemented using Android Studio, with Java as the core programming language, and Firebase was employed for cloud-based data storage and backend management. GPRS technology was integrated to enable real-time vehicle tracking, while a routing optimization algorithm was embedded to reduce travel time and improve delivery accuracy. To evaluate the system, a prototype was tested with 50 participants, including delivery agents and customers, and assessed using metrics such as delivery time, route accuracy, usability, and customer satisfaction. The findings revealed that the application reduced average delivery time by 18%, improved routing accuracy by 25%, and achieved a customer satisfaction rate of 91%. These results demonstrate that integrating GPRS technology within delivery applications can substantially improve efficiency, accessibility, and transparency in service delivery. Furthermore, the proposed system provides a scalable model adaptable to e-commerce platforms and logistics companies, particularly in developing regions, where infrastructure and technological constraints often hinder efficient service provision.

Keywords: Mobile Application; GPRS Technology; Delivery Services; Routing Optimization; Efficiency; Accessibility

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  - **Min** 3 keywords
  - **Max** 6 keywords

Semicolon

DEDICATION

Sample **ACKNOWLEDGMENT**

We extend our deepest gratitude to our supervisor, ***[Mohamed Abdullahi Ali)]***, for his unwavering support, patient guidance, and insightful feedback throughout our research journey, which have been instrumental in shaping the quality and direction of this thesis. We are equally thankful to the Research Coordinator**,** ***Mohamed Abdullahi Mohamud***, and the Assistant Research Coordinator**,** ***Abdullahi Mohamed Hassan***, for their continuous assistance, encouragement, and timely guidance that contributed greatly to the smooth progress of this study.

Our appreciation also goes to the faculty and staff in the Department of **[Bachelor of Computer Applications]** at **Jamhuriya University of Science and Technology**, whose resources and assistance have been invaluable. We would also like to acknowledge our peers for their companionship and the exciting discussions that inspired us throughout our academic journey. Their collective wisdom and encouragement have been a cornerstone of our research experience.

We would like to acknowledge with sincere respect the exemplary leadership of our dean**,** ***Mohamed Abdullahi Ali***, whose vision and encouragement have inspired us to strive for higher standards of academic achievement.

Finally, we are profoundly grateful to the President of Jamhuriya University of Science and Technology**,** ***Mohamed Ahmed Mohamud****,* for his dedication to advancing higher education and research, as his leadership has created opportunities that enabled us to pursue and complete this academic endeavor.

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LIST OF ABBREVIATION

AI	Artificial Intelligence
ANN	Artificial Neural Network
CNN	Convolutional Neural Network
DL	Deep Learning
GPS	Global Positioning System
IoT	Internet of Things
ML	Machine Learning
MLP	Multilayer Perceptron
NDVI	Normalized Difference Vegetation Index
RF	Random Forest
RGB	Red, Green, Blue (color model)
ROI	Region of Interest
SVM	Support Vector Machine
UAV	Unmanned Aerial Vehicle
YOLO	You Only Look Once (real-time object detection algorithm)

CHAPTER I: INTRODUCTION

1.1 Background of study

The Background of the Study is the opening part of Chapter 1 in a research or project report. Its purpose is to introduce the research topic, explain the broader context, and highlight why the study is important. It provides an overview of:

- The global and local trends related to the study.
- The relevance of the chosen technology or solution.
- The justification for conducting the study.
- Existing challenges or gaps around interest.

This section sets the foundation for the problem statement, objectives, and significance of the study that follow.

1.2 Problem Statement

The problem statement is one of the most important parts of Chapter 1 in a research or project report. Its main purpose is to:

1. Define the Problem Clearly: Narrow the broad background into a specific, researchable issue.
2. Justify the Study: Explain why solving this problem is important for users, businesses, or society.
3. Provide Direction: Set the stage for the objectives, research questions, and methodology.

Without a clear problem statement, the research lacks focus and purpose.

A well-structured problem statement usually has four main sections:

1. Ideal Situation (Best Case Scenario)
2. Current Situation (The Problem)
3. Consequences of the Current Situation
4. Research Promise (Proposed Solution)

From ideal → to problem → to consequences → to proposed solutions.

1.3 Research Objectives

The Research Objectives define what your study aims to achieve. They serve as a roadmap for the project and should be:

- Clear – easy to understand.
- Achievable – realistic within the scope of an undergraduate project.
- Specific – focused on solving the identified problem.
- Measurable – so progress and success can be evaluated.

** Must be implementable objectives.

1.4 Research Questions

The Research Questions guide your study by translating your objectives into specific queries that the research seeks to answer.

1.5 Motivation of study

The Motivation of the Study explains the driving reasons behind choosing the research topic. Unlike the *Significance of the Study* (which focuses on *who benefits*), the *Motivation* focuses on *why the researcher chose the problem* and *why it is worth solving*.

It typically includes:

- Personal/academic interest in the topic.
- Practical problems observed in real life.
- The potential to contribute to solutions in society.
- The opportunity to explore technology or innovation.

1.6 Significance of Study

The Significance of the Study explains why the research is important. It shows the potential benefits of the study to different stakeholders. This section answers the question: "Who will benefit from this research and how?"

1.7 Scope of study

Scope of the Study is usually the last section in Chapter 1 (before *Organization of the Study*). It clearly defines the boundaries of your research, what you will cover and what you will not cover. It prevents the study from being “too broad” and makes it manageable. The Scope of the Study should clearly answer three things:

1. Context (What the study is about / focus area)
2. Timeframe (When the study is conducted)
3. Geographical Area (Where the study is conducted)

1.8 Organization of the study

The Organization of the Study (sometimes called “Structure of the Study”) explains how the research/project report is arranged. It provides the reader with a roadmap, showing what each chapter contains and what to expect. This section is usually included at the end of Chapter 1.

It typically includes:

- A summary of each chapter.
- The logical flows from introduction to conclusion.
- How the chapters contribute to achieving the research objectives.

CHAPTER II: LITERATURE REVIEW

2.1 Introduction

The Literature Review is Chapter 2 of your research/project report. Its purpose is to:

1. Review existing studies related to your topic.
2. Identify gaps in previous research or applications.
3. Provide a foundation for your project by showing how it builds on or differs from existing work.

Demonstrate understanding of the field (technologies, theories, methods).

2.2 Overview of ………

2.3 Related Work

The Related Work section reviews past research studies, journal articles, and projects closely connected to your topic. This section emphasizes:

- What previous researchers have done.
- What methods/technology are they used.
- What they achieved.
- Their limitations.
- How your study builds on or differs from theirs.

In Computer Science theses, Related Work often ends with a comparison table and a transition to the research gap. A comparison table is a great way to summarize key studies in your literature review. It makes it easy to see what previous research did, their methods, strengths, limitations, and gaps.

Table 2.1 Comparison of Existing Studies on Delivery Applications and GPS Integration

Author & Year	**Description**	**Methods**	**Limitations**

2.4 Research Gap

The Research Gap highlights:

- What has already been studied about the topic.
- What has not been studied or addressed properly.
- Why solving this gap is important.
- How your project intends to fill this gap.

It is the “bridge” between the literature review and your research objectives.

CHAPTER III: METHODOLOGY

(Mobile and Web Dev FYPs)

3.1 Introduction

This chapter outlines the description of the system, features, architecture, development approach, hardware and software requirements, and user requirements. It provides a structured framework for understanding how the system will achieve the research objectives.

3.2 System Overview

System Overview is usually included in the Methodology chapter (Chapter 3) as a section that summarizes the application you are building. It explains the big picture of the system, without going too deep into code (that comes later in *System Design & Implementation*).

It answers: *What does the system do?*

3.3 System Features

System Features description is usually more detailed than just listing features. It describes what each feature does, why it is important, and how it works in the system.

3.4 System Architecture

The System Architecture section is included in Chapter 3 (Methodology) of a computer applications thesis because it explains how the system is structured, how components interact, and how data flows. Essentially, it shows the blueprint of your solution.

***** Without it, readers may not fully understand how your methodology translates into a functioning application.***

Insert image here →----------

3.5 System Requirements

The System Requirements section outlines the necessary hardware, software, and technical specifications needed to design, develop, and deploy the application. This ensures that the system functions efficiently, is compatible with target devices, and meets user expectations.

3.5.1 Hardware Requirements

The hardware requirements define the physical components necessary for developing, testing, and running the application efficiently. These requirements ensure the system performs optimally and supports all functional and non-functional features.

3.5.2 Software Requirements

The software requirements define the necessary programs, platforms, and tools needed to develop, test, and run the application efficiently. These requirements ensure compatibility, smooth development, and proper functioning of all application features.

- **Development Tools**: ……. (IDE)
- **Programming Languages**: …….
- **Database**: ……………………..
- **Frameworks & Libraries**: …….
- **Operating Systems**: ……………

3.6 Requirements

3.6.1 Functional Requirements

Functional Requirements describe what the system must be able to do. They should list the main functionalities in a clear and structured way so that anyone reading can easily understand the purpose of each function. Each requirement needs to focus on the system's expected behavior, and it should be written in a way that is specific, measurable, and testable. To keep everything organized, requirements should be labeled systematically, for example R1, R2, R3, and so on.

3.6.2 Non-Functional Requirements

Non-Functional Requirements define the qualities and constraints of the system rather than the specific functions it performs. They should be organized into categories such as performance, usability, reliability, security, scalability, portability, and maintainability. Each requirement must be written in clear and precise statements so it can be understood and evaluated. To ensure consistency, non-functional requirements should also be labeled systematically, for example NFR1, NFR2, NFR3, and so on.

3.7 Feasibility Study

A feasibility study assesses whether it is practical to develop the proposed system within the given resources and constraints. It should analyze different aspects to ensure the project can be successfully completed; Covers the following aspects:

- **Technical Feasibility** – tools, platforms, technologies required.
- **Economic Feasibility** – cost-benefit and resource affordability.
- **Operational Feasibility** – user acceptance and ease of adoption.
- **Schedule Feasibility** – time required vs. project deadlines.

3.8 UML Diagrams

All UML diagrams must be created, labelled, and placed in the Appendix section with proper figure numbering. In the chapter text, only reference them.

3.8.1 Use Case Diagram

- Identify system actors and their interactions.
- Show major use cases and relationships between actors and the system.
- Refer to the corresponding diagram in the Appendix (e.g., *Appendix Figure A2: Use Case Diagram*).

3.9 Database Design

- Describe the logical and physical design of the database.
- Define entities, attributes, primary keys, and foreign keys.
- Explain normalization level and justification for design choices.
- Include a clear schema of tables and relationships.
- Refer to Figure templates of database schema placed in the Appendix with proper numbering (e.g., *Appendix Figure 4.x and table 4.1*).

CHAPTER IV: IMPLEMENTATION AND RESULTS
(Mobile and Web Dev based FYPs)

4.1 Introduction

This section provides an overview of the system's implementation, testing, and evaluation. It connects the work done in the previous chapters moving from system design to actual implementation, followed by testing and validation. The purpose of this stage is to ensure that the developed system functions as expected and meets the requirements defined earlier.

4.2 Snapshots of the System

This part presents screenshots of the system to show its main features and how they work. It includes details of the web admin panel, testing, and usability checks.

4.2.1 Web Admin Panel Implementation and Testing

The admin panel is important because it helps administrators manage users, services, reports, and notifications. Each feature supports the system's goals, and screenshots should be used to explain how the modules function.

4.2.1.1 Functional Testing

Functional testing must do using a black-box approach, focusing on inputs and outputs. Key functions to be tested include login, CRUD operations, and data validation. Test cases must be recorded like the input, expected result, actual result, and pass/fail status.

4.2.1.2 UI/UX Evaluation

It should discuss the design principles applied, including consistency, responsiveness, and usability. Comment on navigation, clarity, and system performance, and explain any improvements made based on testing feedback.

4.2.2 Mobile Application Implementation and Testing (If Used Mobile App)

4.2.2.1 Mobile App Features

Start by briefly explaining the purpose of the mobile app and how it complements the main system. List the key features, such as user registration/login, booking or order processes, notifications, and profile management. For each feature, explain how it supports the system objectives and improves convenience for the user. Include screenshots of the main interfaces to illustrate the design and functionality.

4.2.2.2 Unit and Integration Testing

Describe the approach used for unit tests on individual modules and integration tests on combined modules. Provide sample test cases that show the input, expected output, and actual result. Pay special attention to critical areas such as authentication, database communication, and API calls.

4.2.2.3 Device Compatibility Testing

Device compatibility testing to be included to ensure the app works correctly across multiple devices. Explain why testing on different platforms and screen sizes is important. List the devices and operating systems tested (e.g., Android and iOS, various screen sizes, OS versions), and discuss how the app performed in terms of responsiveness, layout consistency, and overall usability across devices.

4.3 Backend and API Development & Testing

4.3.1 Backend

Explain the role and purpose of the backend, including the frameworks and databases used, such as Node.js, Django, or MongoDB. A simple diagram can be included to show the backend architecture and how it interacts with other system components.

4.3.2 API Development (if used)

Describe their purpose in connecting the frontend and backend, and specify the API style, such as RESTful, along with any standards or conventions followed during development

4.3.3 Testing

Testing the backend and APIs is critical to ensure the system is correct, stable, and performs efficiently. Students should explain their testing goals, including checking functionality, handling errors, and performance under load. Tools like Postman can be used to test API endpoints and verify responses. Discuss how error handling was validated, for example with invalid inputs or proper error codes, and summarize any load testing conducted along with the results observed.

4.4 Security Implementation and Testing

4.4.1 Authentication and Authorization

Explain how users are identified and granted appropriate access. Include:

- Authentication methods (e.g., login with email/password, token-based, JWT, OAuth).
- Authorization rules (e.g., admin vs user access, role-based access control).
- Session management or token expiration for protection.

4.2.2 Input Validation

Show how the system defends against malicious input and common attacks. It includes:

- Input validation techniques (client-side and server-side).
- Use of libraries or frameworks that enhance security.

4.2.3 Security Testing Methods

Demonstrate how the system's security was verified. Includes:

- Manual testing or automated tools used (e.g., OWASP ZAP, Burp Suite).
- Penetration testing, vulnerability scanning, or code review activities.
- Summary of discovered issues (if any) and how they were fixed.
- Result: confirm the system meets basic security standards.

CHAPTER V: DISCUSSION AND CONCLUSION

5.1 Introduction

This section summarizes the results of your project and evaluates how well the system met its objectives.

5.2 Discussion

In the discussion, students should interpret the results of their project, explaining what they mean in relation to the system's goals. Discuss whether the system performed as expected, how effective it is in solving the identified problem, and any unexpected findings. Support your discussion with data, screenshots, performance metrics, or user feedback collected during testing.

5.2.1 Comparison with Existing Studies

Compare your system or findings with similar projects, studies, or existing solutions, especially those relevant to Somalia or Mogadishu if possible. Highlight similarities and differences, and explain why your approach is better, different, or more suitable for the local context. For example, if you developed mobile registration app, compare its features, performance, or usability with other local or international apps.

5.3 Conclusion

The conclusion summarizes the key results of your project and presents a final statement on the system's success. It should be concise, showing the overall outcomes, and reinforcing the significance of your work.

5.3.1 Achievement of the Objectives

List each research objective in chapter 1 and state whether it was fully, partially, or not achieved. Provide evidence from the system implementation, testing, or evaluation to support your claim. This shows supervisors and examiners that you systematically addressed the project goals.

5.4 Limitation

Discuss the limitations or challenges faced during the project. These can include technical issues, time constraints, lack of resources, or scope restrictions. Being honest about limitations demonstrates critical thinking and awareness of real-world project constraints.

5.5 Recommendations

Provide practical suggestions for improving the system or extending its functionality in the future. Recommendations can include new features, performance enhancements, security improvements, or broader deployment. Future work may also suggest research directions or new technologies that could be applied to similar projects in Mogadishu or other contexts.

REFERENCES (APA 7th / IEEE Style)

Correct referencing is a vital part of academic writing. It shows respect for the work of others, strengthens your arguments, and prevents plagiarism. At JUST, two styles of citation are recommended:

- **APA 7th Edition:** For graduation projects, dissertations, and theses.
- **IEEE Style:** For conference papers, journal articles, and research publications.

Students must use the correct style consistently, depending on the purpose of their work.

6.1 APA 7th Edition (for Thesis/Report Writing)

In-Text Citation Examples:

- Paraphrase: *(Khalaf et al., 2025)*
- ***Example:***
- Direct quote: *"SMA appears to harm academic performance through two distinct routes: a direct pathway linked to motivational deficits and self-regulation failure, and an indirect pathway via energy depletion from poor sleep." (Khalaf et al., 2025, p. 1).*

Reference List Examples:

Khalaf, M. A. A., Omar, A. I., Mohamud, M. A., & Abdulle, A. W. *Social Media Addiction and Academic Engagement: The Role of Sleep Quality and Fatigue among University Students in Somalia*. In Frontiers in Education (Vol. 10, p. 1614746). Frontiers.

A. A. Adam, I. A. Farah, M. A. Yusuf, B. A. Hussien and A. A. Siyad, "*Optimizing White Blood Cell Nuclei Segmentation: A Proficient Method for Improved Diagnostic Accuracy*," 2024 IEEE International Conference on Data and Software Engineering (ICoDSE), Gorontalo, Indonesia, 2024, pp. 188-193, doi: 10.1109/ICoDSE63307.2024.10829884.

6.2 IEEE Style (for Research Papers/Publications)

In-Text Citation Examples:

- Single reference: *[1]*
- Multiple: *[1], [2]*
- Example Usage: The SFF, like many other federations, faces challenges in collecting, managing, and utilizing football statistics effectively [1].

Reference List Examples:

[1] Ahmed, B. A., Hashi, H. A., Ahmed, A. A., & Ali, A. M. (2025). Developing an Integrated Platform to Track Real Time Football Statistics for Somali Football Federation (SFF). International Journal of Advanced Computer Science & Applications, 16(1).

7.1 Document Structure

Your final report must follow the official order:

Preliminary Pages:	Main Text:	Supplementary:
Title Page, Declaration, Abstract, Acknowledgement, Table of Contents, List of Figures/Tables/Abbreviations, Appendices list (if any).	Introduction, Literature Review, Methodology, Implementation & Results, Discussion & Conclusion.	Appendices, Bibliography/References (APA 7th or IEEE).

7.2 Length

The main report (excluding appendices, tables, and prefaces) should be 15,000–20,000 words.

7.3 Formatting Rules

- **Font-family & Size**: Times New Roman, size 12.
- **Line Spacing**: 2.0 for text, single-spacing for tables, references, and long quotations.
- **Margins**: Left: 4.0 cm, Top: 2.0 cm, Right: 2.0 cm, Bottom 2.0 cm.
- Paragraphs of the normal text will be aligned between the extreme left and right (**justify**).
- **Printing**: Use clear black text on A4 (80g) white paper.
- **Binding**: Hardbound copies must use approved faculty colors of dark red. gold font (size 18) for cover text.

7.4 Page Numbering

All page numbers should be printed 1.0 cm from the bottom margin and placed at the **right-hand** side without any punctuation.

- Roman numerals (i, ii, iii etc) should be used in the Preface section. Although the Title Page is the first page of the Preface, no number is printed on it. Numbering begins on the second page with (ii).
- Arabic numerals (1, 2, 3) are used on the pages of the text (starting with the introduction page) and supplementary sections.
- Font size 8 recommended for numbers.

7.5 Tables and Figures

Tables and figures must be centered on and numbered by chapter (e.g., *Table 3.2, Figure 4.5*).

Titles:

- The tables label should be placed **above** the table itself.
- The figures labels are placed at the **bottom** of the figure.

If the figure occupies more than one page, the continued figure on the following page should indicate that it is a continuation: for example: Figure 3.7, continued'.

If the figure contains a citation, the source of the reference should be placed at the bottom, after the label.

7.6 Referencing

Use **APA** 7th edition or IEEE style, consistently throughout.

In-text citation examples:

APA: *(Ali, 2022)*

IEEE: *[3]*

7.8 Appendices

Place supporting material (questionnaires, code, raw data, maps, screenshots) in Appendices.

Label as Appendix A, B, C, etc.

7.9 Turnitin Similarity & AI Detection

All Final Year Project (FYP) reports must comply with the highest standards of academic honesty. To ensure originality, submissions will be screened using Turnitin similarity check and, where necessary, AI-generated content detection tools.

- **Turnitin Similarity Check:**
    - ❖ Every FYP must be checked in Turnitin before submission.
    - ❖ The acceptable similarity index is **20% or less**.
    - ❖ Any highlighted matches must be properly paraphrased or cited.
    - ❖ Reports with high similarity will be returned for correction or may be rejected.
- **AI-Generated Content and Detection:**
    - ❖ Students may use AI tools for supportive purposes such as idea generation, grammar checking, or formatting references.
    - ❖ Submitting AI-generated content as original work is strictly prohibited.
    - ❖ AI-detection software may be used to identify excessive reliance on AI writing tools.
    - ❖ Students must demonstrate ownership of their work during the viva by explaining their methodology, design, and results in detail.

8.1 Introduction

The presentation and defense allow students to demonstrate their project work, justify their design decisions, and answer questions from a panel of examiners. The panel assesses the technical quality, research contribution, clarity of communication, and the student's confidence and understanding.

8.2 Slide and System Presentation Structure (Suggested 15–20 Minutes)

Students should prepare a clear, concise, and professional slide deck covering the following:

1. Title Slide – Project title, student name, supervisor's name, department.
2. Introduction – Background, problem statement, and motivation for the project.
3. Objectives – Main aims and expected outcomes of the project.
4. Literature Review (Brief) – Existing solutions and gaps your project addresses.
5. Methodology – Approach, tools, architecture, and design (diagrams recommended).
6. Implementation – Hardware/software components, screenshots, photos, or demos.
7. Results – Tables, graphs, performance metrics, user feedback, or evaluations.
8. Discussion – Interpretation of results, challenges faced, and how they were solved.
9. Conclusion – Achievements, limitations, and recommendations for future work.\
10. System Demo – Show the system prototype or working features.

3. Presentation Tips

- Keep slides simple, visual, and not overloaded with text.
- Use diagrams, photos, charts, and demos rather than paragraphs.
- Practice timing — aim for 10 minutes, leaving 5 minutes for Q&A.
- Dress professionally and maintain eye contact with the panel.
- Anticipate common questions and rehearse your answers.